

```

cv.WaitKey(5000)
cv.DestroyWindow("Original Image")
hc = cv.Load("/home/jayneil/haarcascade_frontalface_default.xml")
img = cv.LoadImage("/home/jayneil/beautiful-faces.jpg", 0)
faces = cv.HaarDetectObjects(img, hc, cv.CreateMemStorage())
for (x, y, w, h), n in faces:
    cv.Rectangle(img, (x, y), (x+w, y+h), 255)
cv.SaveImage("faces_detected.jpg", img)
print(faces)
dst=cv.LoadImage("faces_detected.jpg")
cv.NamedWindow("Face Detected", cv.CV_WINDOW_AUTOSIZE)
cv.ShowImage("Face Detected", dst)
cv.WaitKey(5000)
cv.DestroyWindow("Face Detected")

```

Run the code. The source image I used is shown in Figure 6, and the output in Figure 7. In the ninth line, I have stored the Haar classifier that I want to use, in a variable. In the 11th line, I have used the *HaarDetectObjects* function to detect the features I want. This function has three parameters. The first one is the source image, the second the Haar classifier I want to use and the third one creates a memory storage. Then I have used a *for* loop to draw a rectangle around the portion that I have identified in the previous stage, using the *Rectangle*

function, which takes four parameters: the image on which to draw, followed by the two opposite corners of the rectangle, and the last one denotes the colour.

In the third and final part of the series, I will cover how to program and use the Arduino. I will also cover how to use the Pyserial module and how to integrate all the technologies we have learned so far to build an embedded system for image processing.

Please refer to the OpenCV documentation (<http://opencv.itseez.com/>) for more information. 

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The author is a FOSS advocate and loves to explore different open source technologies. His areas of interest include OpenCV, Python, Android, Linux, Arduino and other open source hardware platforms. In his spare time, he likes to make video tutorials on various open source technologies, which can be found on YouTube. He is a big-time Arsenal fan and can be reached at jayneil.dalal@gmail.com.

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